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PROGRAMME: BBA. COURSE DEVELOPMENTAL ECONOMICS
MODE OF STUDY: ODL 2023.

CALCULATING THE GINI COEFFICIENT = 1 - Total Score

Fraction of Income		Fraction of population		Fraction of population Richer.	Fraction of Income X (Fraction of population +2 X Fraction of pop. richer)
% Income	Cummulative %	% Pop. lowest	% Cummulative		= Score.
0.1	10%	0.25	25%	0.75	0.175
0.15	25%	0.25	50%	0.50	0.1875
0.35	60%	0.25	75%	0.25	0.2625
0.40	100%	0.25		0.00	0.1
Total.				Total Score	0.725.

Table 1.

The GINI Coefficient = 1 - 0.725

= 0.275

COMMENTS BASED ON THE ANSWER FOR THE GINI COEFFICIENT

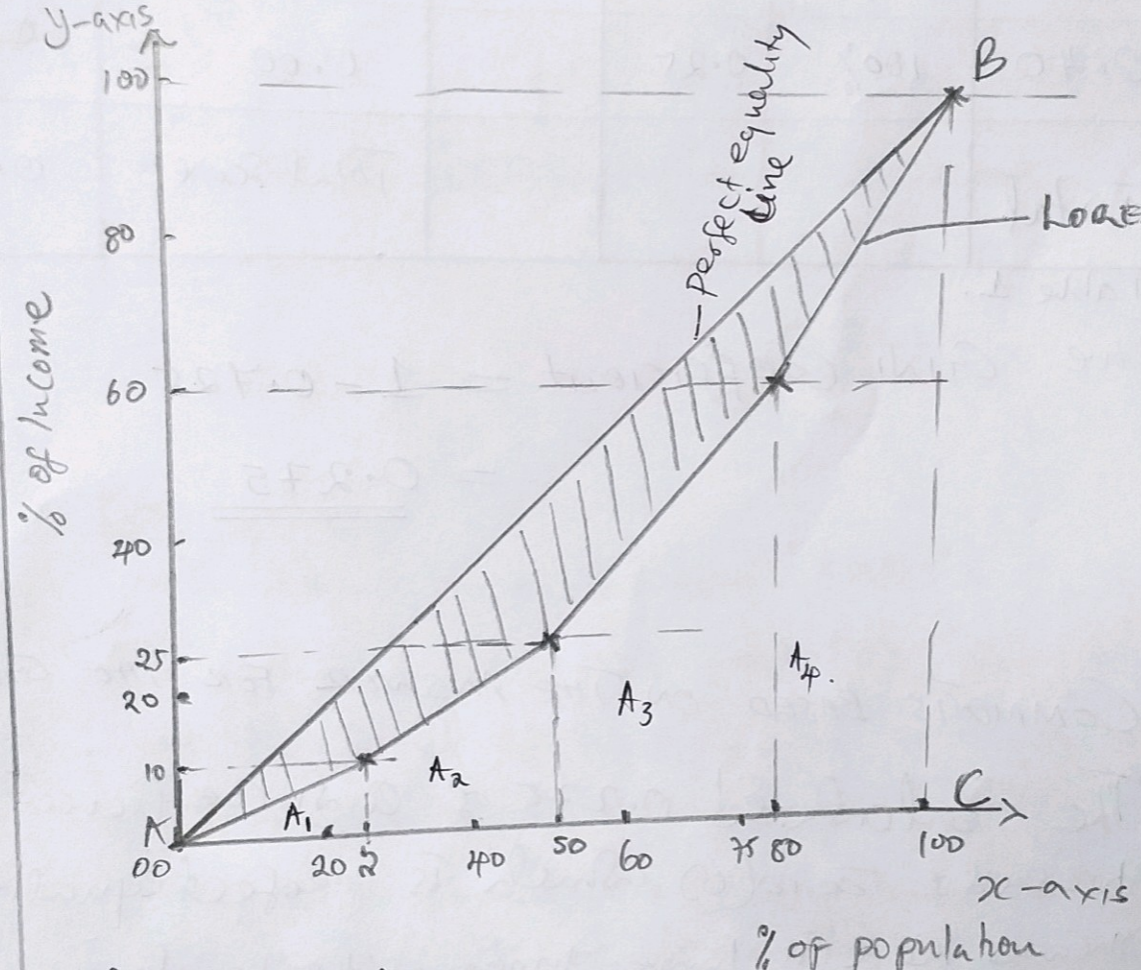
The Calculated 0.275 of GINI Coefficient is going towards Zero(0) which is perfect equality, it means there is lower inequalities levels in this Country as the gap interms of Income between the rich and poor is lower. This means the Country is going towards Development using this measure. GINI Coefficient of 0.275 means there is relative equality.

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Q2 SECOND METHOD.

CALCULATING THE GINI COEFFICIENT USING THE LORENCE CURVE.

Ploting The % of Income Cummulative numbers and percentage of population lowest Cummulative figures as x-axis. Using a Scale of 1:20 in both ~~Axe~~-axises. See Table 1



Calculating total Area of Triangle = $\frac{1}{2}AB = \frac{1}{2} \times 100 \times 100 = 5000$ units.

$A_1 = \frac{1}{2} \times 10 \times 25 = 125$ units

$A_2 = \frac{1}{2}(a+b)h = \frac{1}{2}(10+25) \times 25 = 437.5$ units.

$A_3 = \frac{1}{2}(a+b)h = \frac{1}{2}(25+60) \times 25 = 1062.5$ units

$A_4 = \frac{1}{2}(a+b)h = \frac{1}{2}(100+60)25 = 2000$ units

ADDING $A_1 + A_2 + A_3 + A_4 = 125 + 437.5 + 1062.5 + 2000$

$= 3625$ units.

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b) GINI COEFFICIENT USING THE LORENZ CURVE

$= 1 - \frac{\text{AREA UNDER THE LORENZ CURVE}}{\text{TOTAL AREA OF TRIANGLE ABC}}$ Or $\frac{\text{Total Area} - \text{Area under Lorenz}}{\text{Total Area}}$

$\Rightarrow \frac{3625 \text{ units}}{5000 \text{ units}} = 0.725$ Or $\frac{5000 - 3625}{5000}$

$\Rightarrow = 0.275$

∴ GINI COEFFICIENT $= 1 -$ SCORE:

$= 1 - 0.725$

$= 0.275$

Comments

Both answer in method 1 and method 2 is 0.275. This value is closes to Zero (0). This implies That the level of inequality is lower. The standard is that as you go towards 1 the inequality levels are high and that shows underdevelopment of the nation. As you go Towards 0. It shows Development as there is ~~equality~~ of Income is Shared or spread equally between the poor and the rich.

0.275 means 27.5% of